

AWS-hosted Virtual Classroom and Learning Platform

Prepared For

Smart-Internz
Cloud Practitioner

By

Bhaskar Jaysing Paymal

D Y Patil Agriculture and Technical University, Talsande

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Project Title:

AWS-hosted Virtual Classroom and Learning Platform

Category:

Cloud Deployment | Web Application | AWS Cloud Practitioner

Skills Utilized:

- Core Python
- Flask Framework
- AWS EC2, S3, RDS
- MySQL
- HTML, CSS, JavaScript
- Git & GitHub

1. Project Overview:

In the modern era of digital learning, building a secure, scalable, and flexible virtual classroom is crucial. This project demonstrates the integration of Flask and multiple AWS services to develop a cloud-native educational platform.

Deployed on AWS EC2, the application leverages Amazon S3 for content storage and Amazon RDS (MySQL) for user and application data. Users can register, log in, and access course materials hosted on the cloud — making the platform both effective and expandable for future needs.

2. Key Features:

- Scalable Infrastructure using EC2
- Secure File Storage using S3
- User Management via RDS (MySQL)
- Responsive Web UI for students and instructors
- Seamless Cloud Integration with GitHub

3. Architecture:

```
project/
├─ app.py
├─ templates/
│   ├─ home.html
│   ├─ register.html
│   ├─ login.html
│   └─ content.html
└─ static/ # If needed for CSS, JS, or images
```

4. Final Project Flow:

4.1 Create an AWS Account:

- Sign up and verify your account.
- Explore the AWS Management Console.

4.2 Create an S3 Bucket and Upload Data:

- Create a bucket (e.g., Aws-classroom-content).
- Upload files (PDFs, videos).
- Set proper permissions (public or signed URLs).

4.3 Create an RDS Instance (MySQL):

- Launch RDS with MySQL engine.
- Configure DB instance and create a database.
- Connect using MySQL Workbench to create tables.

4.4 Launch and Configure EC2 Instance:

- Launch instance with Amazon Linux 2 or Ubuntu.
- Set security groups and SSH key pair.
- Install Python, Flask, MySQL client.

4.5 Develop Flask App:

- Build routes for register, login, content.
- Create templates: `home.html`, `register.html`, `login.html`, `content.html`.
- Use Bootstrap for styling.
- Connect app to AWS S3 (using `boto3`) and RDS.

4.6 Deploy Flask App on EC2:

- SSH into EC2.
- Clone GitHub repository.
- Install dependencies: `pip install -r requirements.txt`.
- Run app using Gunicorn + Nginx (optional).

4.7 Upload Code to GitHub:

- Create repository.
- Push project files with commits and documentation.

4.8 Test Scenarios:

- **Scenario 1: Student Registration and Login**
- **Scenario 2: Admin Upload of Course Materials**

- **Scenario 3: Downloading Course Content**

5. User Scenarios:

Scenario 1: Student Registration and Course Access

- **User:** Alice Johnson
- **Process:** Registers via form, logs in, and accesses course content from S3.

Scenario 2: Admin Uploads Course Materials

- **User:** System Admin
- **Process:** Uploads PDFs; content is stored in S3 and metadata is updated in RDS.

Scenario 3: Student Downloading Course Content

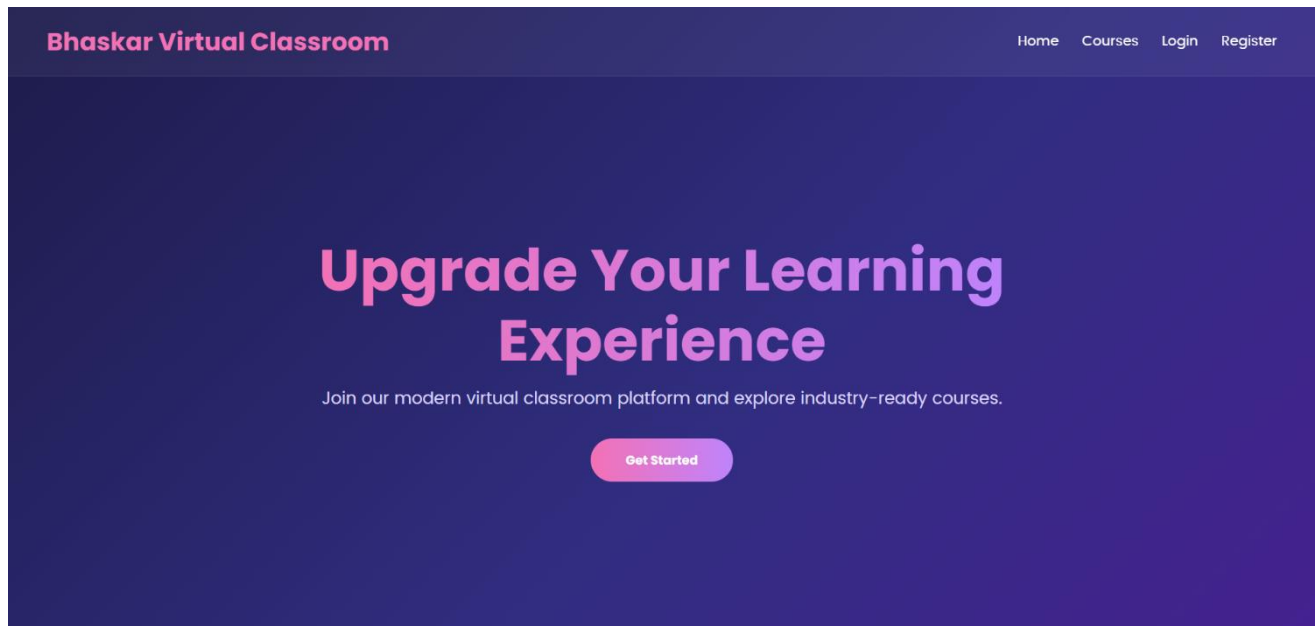
- **User:** Bob Patel
- **Process:** Selects a file, clicks a link, and downloads directly from S3.

6. Challenges Faced:

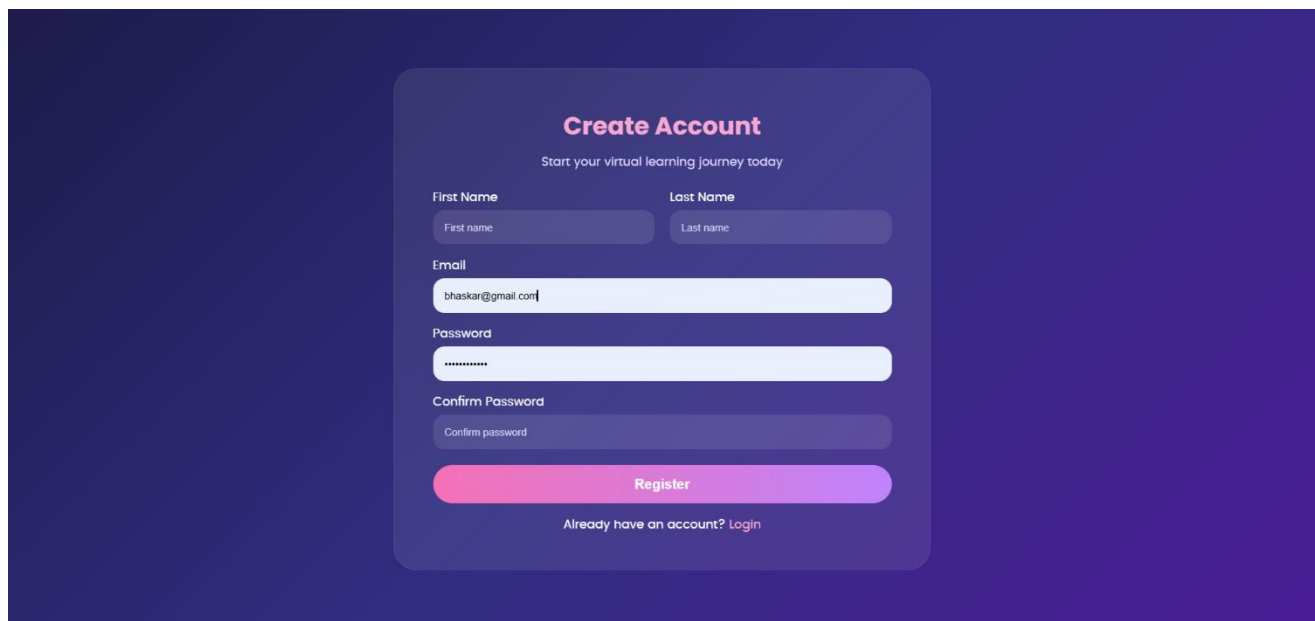
- Learning AWS services and IAM policies
- Managing AWS credentials securely
- Flask and AWS integration (using `boto3`)
- RDS connection issues and MySQL Workbench setup
- Debugging EC2 deployment issues

7. Output Pages:

Landing page with navigation

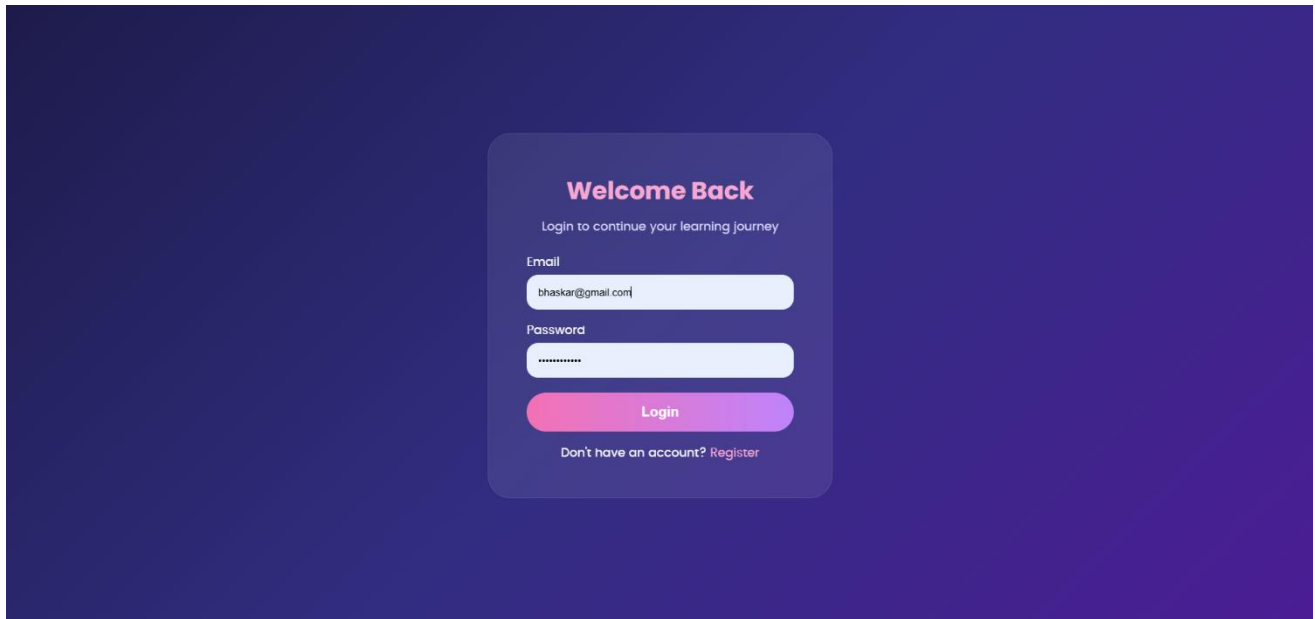


Registration form



The registration form is titled "Create Account" and is set against a dark blue background. It includes the subtext "Start your virtual learning journey today". The form fields are arranged vertically: "First Name" and "Last Name" (each with a placeholder "First name" or "Last name"), "Email" (with placeholder "bhaskar@gmail.com"), "Password" (with placeholder "*****"), and "Confirm Password" (with placeholder "Confirm password"). A light blue rounded button labeled "Register" is positioned below the password fields. At the bottom of the form, there is a link: "Already have an account? Login".

Login page:

A login page with a dark purple gradient background. In the center is a white rounded rectangle containing the login form. The form has a heading 'Welcome Back' in pink, followed by the text 'Login to continue your learning journey'. There are two input fields: 'Email' with the value 'bhaskar@gmail.com' and 'Password' with masked characters. A pink 'Login' button is below the fields. At the bottom, there is a link 'Don't have an account? Register' in white.

Welcome Back

Login to continue your learning journey

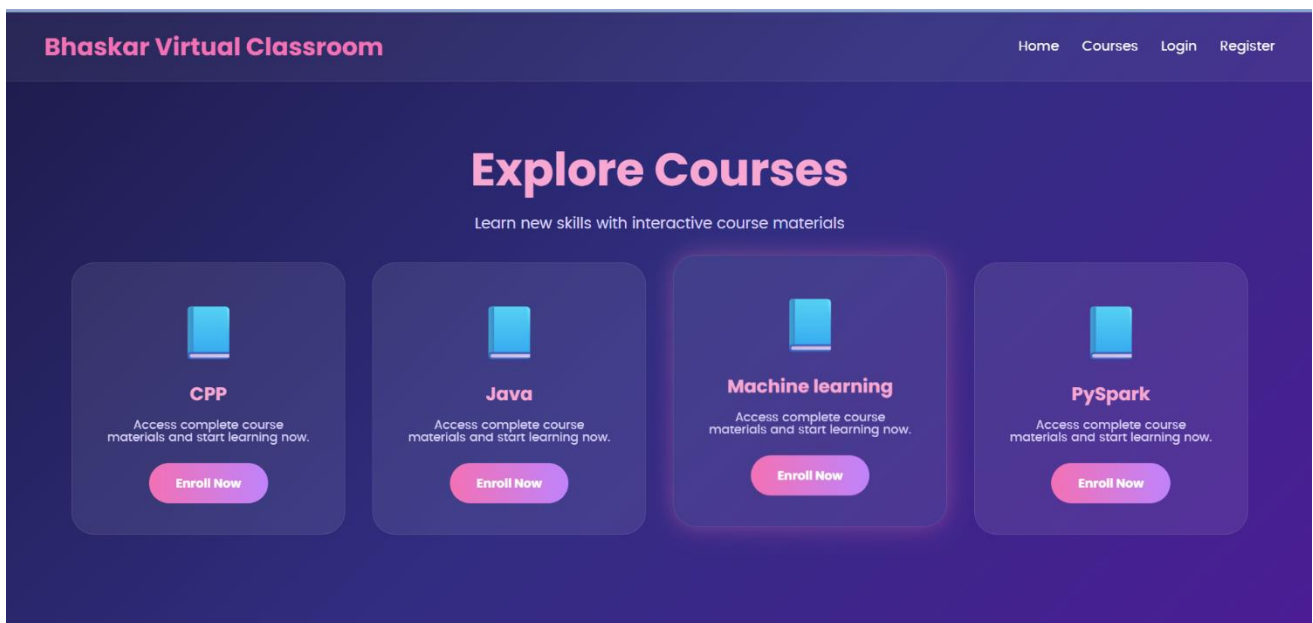
Email

Password

Login

Don't have an account? [Register](#)

Course materials page with download links from S3:

A course materials page with a dark purple gradient background. At the top is a navigation bar with 'Bhaskar Virtual Classroom' on the left and 'Home Courses Login Register' on the right. The main section is titled 'Explore Courses' in pink, with the subtitle 'Learn new skills with interactive course materials'. Below this are four course cards: 'CPP', 'Java', 'Machine learning', and 'PySpark'. Each card has a blue book icon, the course name, a description 'Access complete course materials and start learning now.', and a pink 'Enroll Now' button.

Bhaskar Virtual Classroom Home Courses Login Register

Explore Courses

Learn new skills with interactive course materials



CPP

Access complete course materials and start learning now.

Enroll Now



Java

Access complete course materials and start learning now.

Enroll Now



Machine learning

Access complete course materials and start learning now.

Enroll Now



PySpark

Access complete course materials and start learning now.

Enroll Now

8. Conclusion:

This project highlights the integration of cloud computing with web development to build a fully operational virtual classroom. Leveraging AWS's scalability and Flask's simplicity, the platform achieves reliable user access, secure data handling, and an overall smooth educational experience.

9. GitHub And Demo Link:

Demo Link:

https://drive.google.com/file/d/1Ype_dllwboTtSh1pTE5Ax1EfnfqIRZfx/view?usp=sharing

GitHub Link: <https://github.com/36bhaskar2004/Aws-hosted-virtual-classroom-and-learning-platform.git>